

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
12.	(a)	(i)		add aqueous NaOH and then (neutralise with) HNO ₃ (1) add aqueous AgNO ₃ (1) yellow precipitate for iodine, cream for bromine and white for chlorine (1)	3			3		3
		(ii)		Y is C ₄ H ₉ Cl / chlorobutane (1) award (1) for any explanation supported by a simple calculation e.g. showing that chlorine accounts for 38% of mass of chlorobutane showing that bromine (and iodine) account for greater than 40% of mass of their respective halogenoalkanes showing that the hydrocarbon chain has a mass of 57 and only chlorine has an A _r value less than 40% of the total mass			2	2		
	(b)	(i)		diagram showing $\delta+$ and $\delta-$ (1) lone pair on OH⁻ (1) curly arrows from lone pair and to break C—Cl bond (1) propan-1-ol and Cl⁻ formed (1)	4			4		